

NSE Stock Close Price Prediction using Deep Learning Techniques

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Abstract

This research paper aims to predict the close price of four prominent stocks in the National Stock Exchange (NSE) of India - Hindustan Unilever Ltd (NSE: HINDUNILVR), Infosys Ltd (NSE: INFY), Asian Paints Ltd (NSE: ASIANPAINT), and Hero Motocorp Ltd (NSE: HEROMOTOCO)—utilizing deep learning techniques, specifically Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU). The predictive accuracy of these deep learning models is compared against the baseline performance of linear regression. Given India's position as the fourth-largest stock market globally with a market capitalization of \$4.33 trillion and an ever-growing number of retail investors—anticipated to surpass 25 crores by 2030—the application of artificial intelligence and machine learning in stock market forecasting becomes increasingly significant. By employing advanced deep learning models, this research seeks to provide insights into the potential of artificial intelligence in enhancing stock price prediction accuracy and the implications of such advancements for investors and the broader Indian stock market.

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1. Introduction

The stock market serves as a dynamic and volatile platform where shares of public companies are traded, reflecting the complex interplay of various economic, financial, and psychological factors. The Indian stock market, ranked as the fourth largest in the world, plays a pivotal role in the global financial landscape with a market capitalization of \$4.33 trillion [1]. As the number of DEMAT accounts continues to increase steadily, reaching 13.9 crore as of December 2023 [2], the Indian stock market is witnessing a surge in retail investor participation, highlighting the growing significance of the market.

Stock market prediction has long been a captivating area of research, fueled by the potential financial gains of accurate forecasting. Deep Learning (ML) has become a powerful force in this domain, offering the ability to uncover complex patterns and trends in massive volumes of financial data.

Historically, Classic methods like linear regression and ARIMA (Autoregressive Integrated Moving Average) have served as a foundation in stock price prediction. Their simplicity and interpretability are their advantages, though they often struggle with the non-linearity and volatility inherent in stock markets [3].

In the realm of Machine Learning, SVMs excel in handling high-dimensional data and can model non-linear relationships. They've been successfully employed for stock prediction tasks [4]. However, their sensitivity to hyperparameter selection and challenges in dealing with large datasets are potential drawbacks. Another method used to predict the price of a stock is using Random Forests and Decision Trees. These methods offer interpretability and robustness to outliers. Random forests, as ensembles of decision trees, often outperform single trees in predictive power [5]. Nonetheless, they can overfit noisy financial data.

ANNs, inspired by biological neural networks, can learn complex non-linear mappings between inputs and stock prices. Their ability to automatically extract features is a strength, though they risk overfitting and lack transparency [6].

LSTM is a type of recurrent neural network (RNN) architecture that is well-suited for sequence prediction problems due to its ability to capture long-range dependencies in the data (Hochreiter & Schmidhuber, 1997). In the context of stock market prediction, LSTM has been widely employed to forecast stock prices by learning from historical price data and identifying intricate patterns and trends.

Similarly, Zhang et al. conducted a comparative study of various machine learning algorithms for stock price prediction, including LSTM. The study utilized historical stock price data and technical indicators as input features and found that LSTM exhibited superior performance in predicting stock prices compared to other machine learning algorithms [7].

In this research, we focus on leveraging the capabilities of LSTM and GRU to predict the close prices of three prominent stocks listed on the National Stock Exchange (NSE) of India—Hindustan Unilever, Infosys Limited, and Hero Motocorp Ltd. The predictive performance of these advanced deep learning models will be compared with the baseline performance of linear regression to assess their efficacy in enhancing stock price prediction accuracy in the context of the dynamic and rapidly growing Indian stock market.

Deep learning models, including LSTM and GRU, serve as powerful function approximators, capable of examining complex input-output relationships even in intricate datasets.

LSTM and GRU are specialized variants of recurrent neural networks (RNNs) designed to handle the vanishing and exploding gradient problems encountered in traditional RNNs. LSTM incorporates

memory cells, input gates, forget gates, and output gates, enabling it to effectively capture and remember long-term dependencies in time series data. Similarly, GRU utilizes update and reset gates to regulate the flow of information through the network, facilitating the capture of temporal dependencies.

2. Methodology

2.1. Description of Data

The historical data for the five companies has been collected from Yahoo Finance [23]. The dataset includes 20-year data from 1 January 2000 to 08 March 2024 of Hindustan Uniliver Ltd, Infosys Ltd, Asian Paints Ltd, Pfizer and Hero Motocorp Ltd and Co. The data contains information about the stock such as High Price, Low Price, Open Price, Close Price, Adjusted close Price and Volume on a particular trading day. Only the day-wise closing price of the stock has been extracted. Table 2 shows statistics of the dataset that is used for training and testing.

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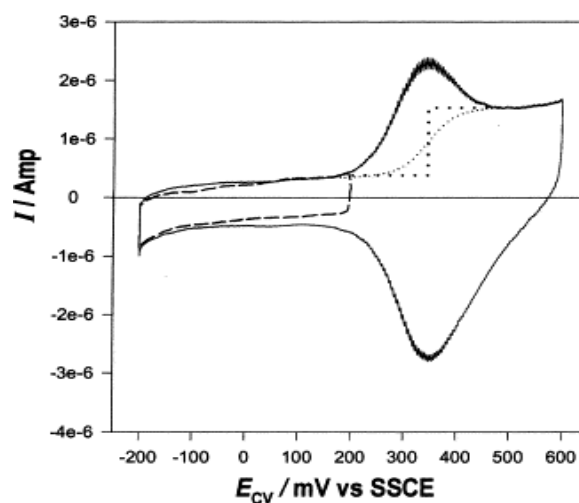


Fig. 1. A caption is positioned left-justified below the figure or scheme.

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Table 1

Summary of data obtained from least-squares fit of Eq. (20) to k_m vs E_i plots for three selected systems

Redox moiety	Diluent	Method	k_0 (s^{-1})	$E_i^{0'}$ V vs SSCE	γ	fwhm (V) ^a
R1	D1	ILIT	3.4×10^4	0.495	95	-
		CV	3.3×10^4	0.474	100	0.103
R2	D2	ILIT	6.0×10^4	0.340	24	-
		CV	6.1×10^4	0.346	24	0.112
R3	D3	ILIT	3.2×10^6	0.328	12.0	-
		CV	3.2×10^6	0.324	12.1	0.121

^aFull-width-half-maximum of the cyclic voltammetric peak.^bThis is the format for table footnotes. Very often you will already have prepared (parts of) your text. If you load that text as a separate document, you can easily insert it into a document based on this template by cutting and pasting between the two documents.

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2. The second entry
 - 2.1. A subentry
3. The last entry
 - A bulleted list item
 - Another one

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